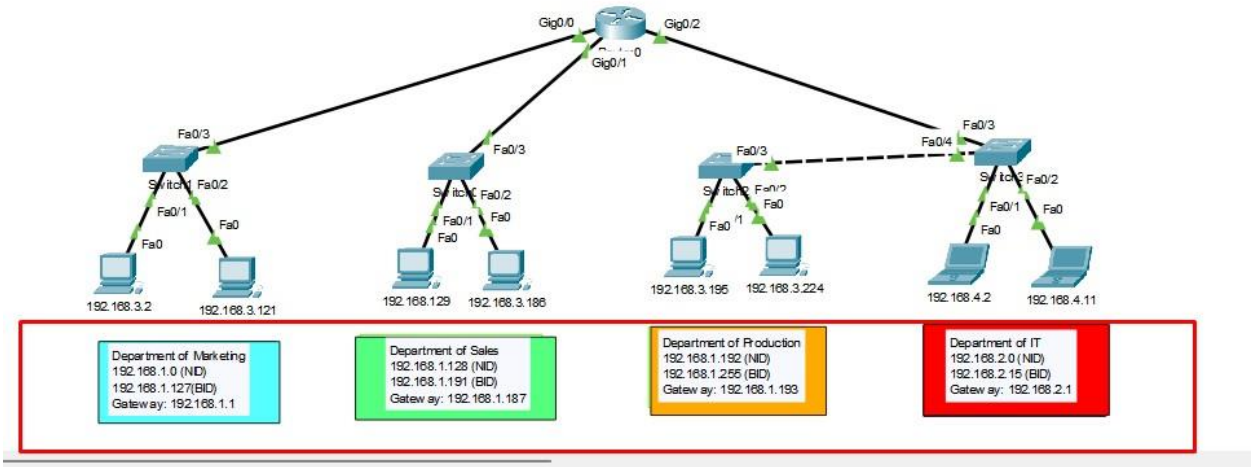
MIKROTIK CONFIGURE

VLSM, DHCP, Simple Queue and NAT

Document overview

The purpose of this document is to assign IP addresses to 4 departments of Company X by subnetting the 192.168.3.0 and 192.168.4.0 networks using VLSM (Variable Length Subnet Masking). This plan prevents IP address waste and is designed for efficient network management. The departments require hosts: 120, 30, 58, 10.

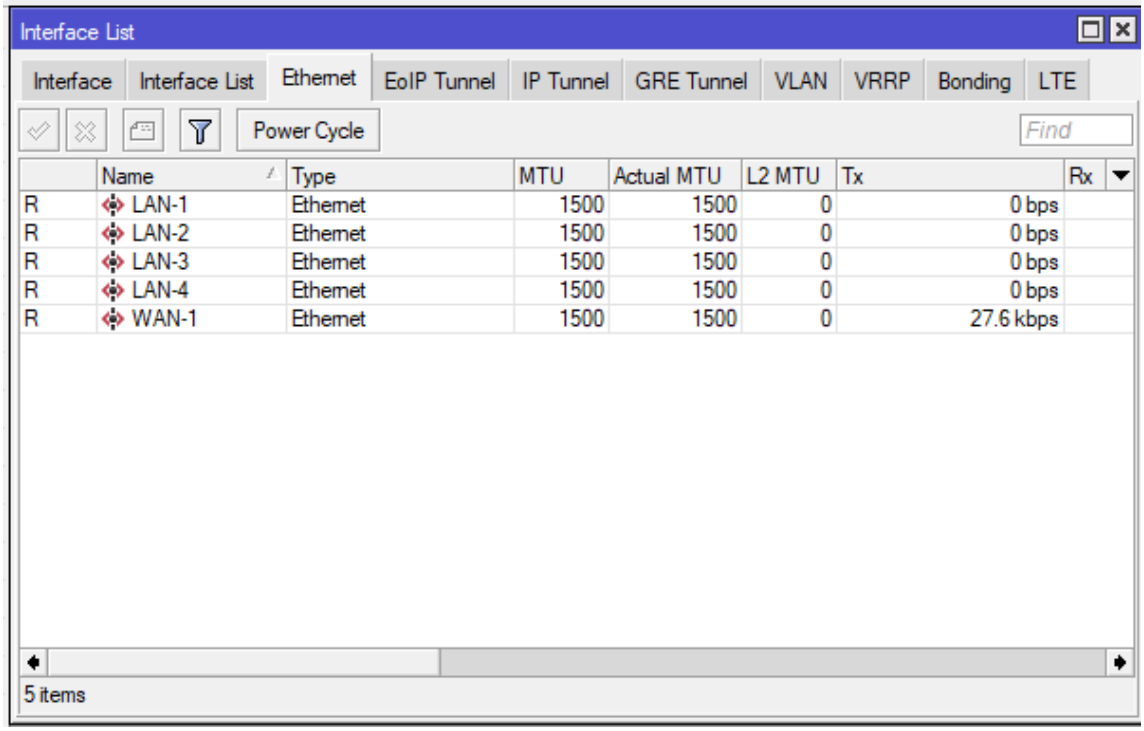


AB Limited company has four departments, there are:
(i) Marketing department: **120** hosts
(ii) Sales department: **58** hosts
(iii) Production department: **30** hosts
(iv) IT department: **10** hosts

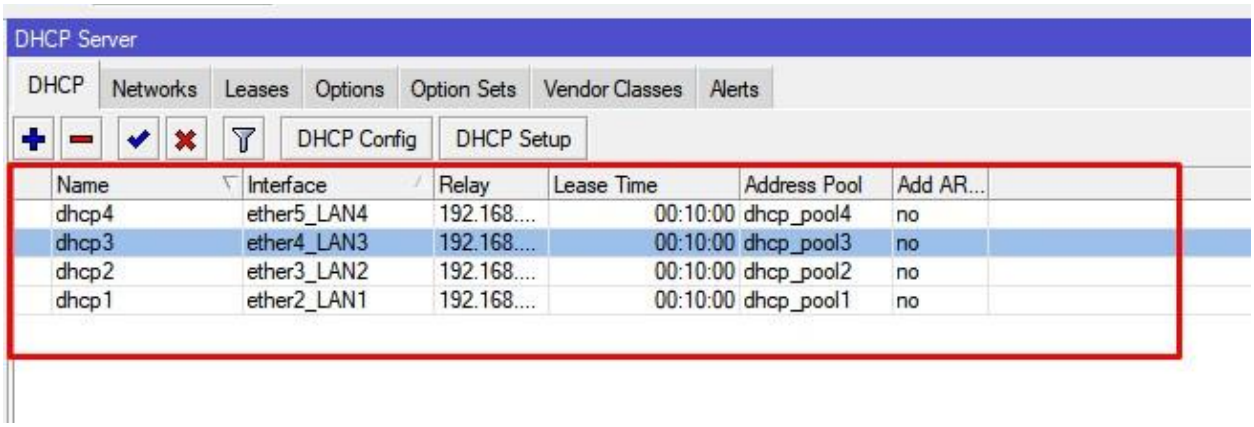
Name of Department	Number of Host	Block Size	S/N	Subnet Mask	Address Range
Marketing	120	128	/25	.128	192.168.3.0 - .127
Sales	58	64	/26	.192	192.168.3.128 - .191
Production	30	64	/26	.192	192.168.3.192 - .255
IT	10	16	/28	.240	192.168.4.0 - .15

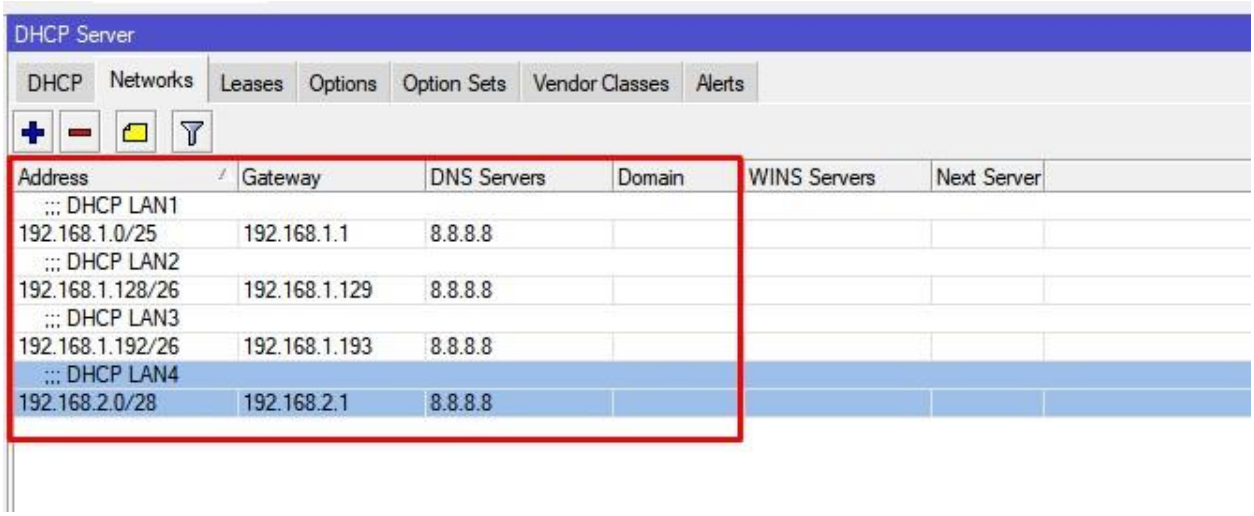
	Name	Type	Actual MTU	L2 MTU	Tx	Rx	Tx Packet (p/s)	Rx Packet (p/s)	FP Tx	FP Rx	FP Tx Packet (p/s)	FP Rx Pa
R	ether1_WAN	Ethernet	1500		56.6 kbps	88.5 kbps	1	87	0 bps	0 bps	0	
R	ether2_LAN1	Ethernet	1500		0 bps	0 bps		0	0 bps	0 bps	0	
R	ether3_LAN2	Ethernet	1500		0 bps	0 bps		0	0 bps	0 bps	0	
R	ether4_LAN3	Ethernet	1500		0 bps	0 bps		0	0 bps	0 bps	0	
R	ether5_LAN4	Ethernet	1500		0 bps	0 bps		0	0 bps	0 bps	0	

Address List: WAN Network-192.168.3.0, LAN-1 Network:192.168.1.0, LAN-2 Network: 192.168.1.128, LAN-3 Network:192.168.1.192, , LAN-4 Network:192.168.2.0

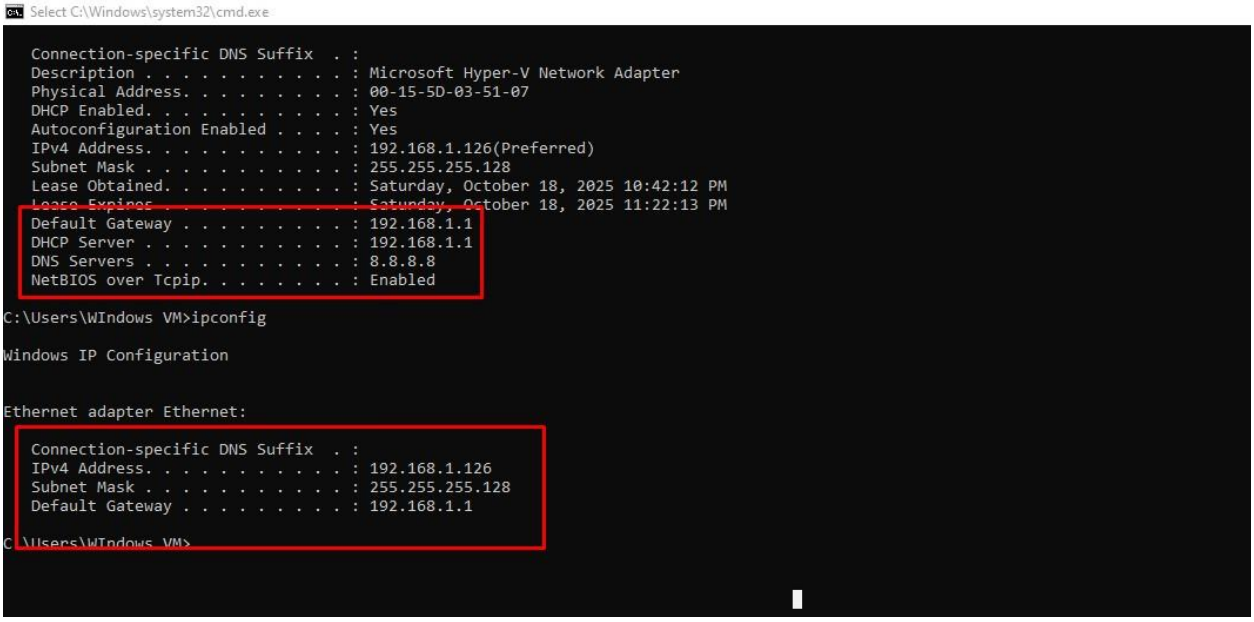


Interface List: WAN-1 Interface connected with ISP. LAN 1,2,3 and 4 Interface connected with Four Different LAN.

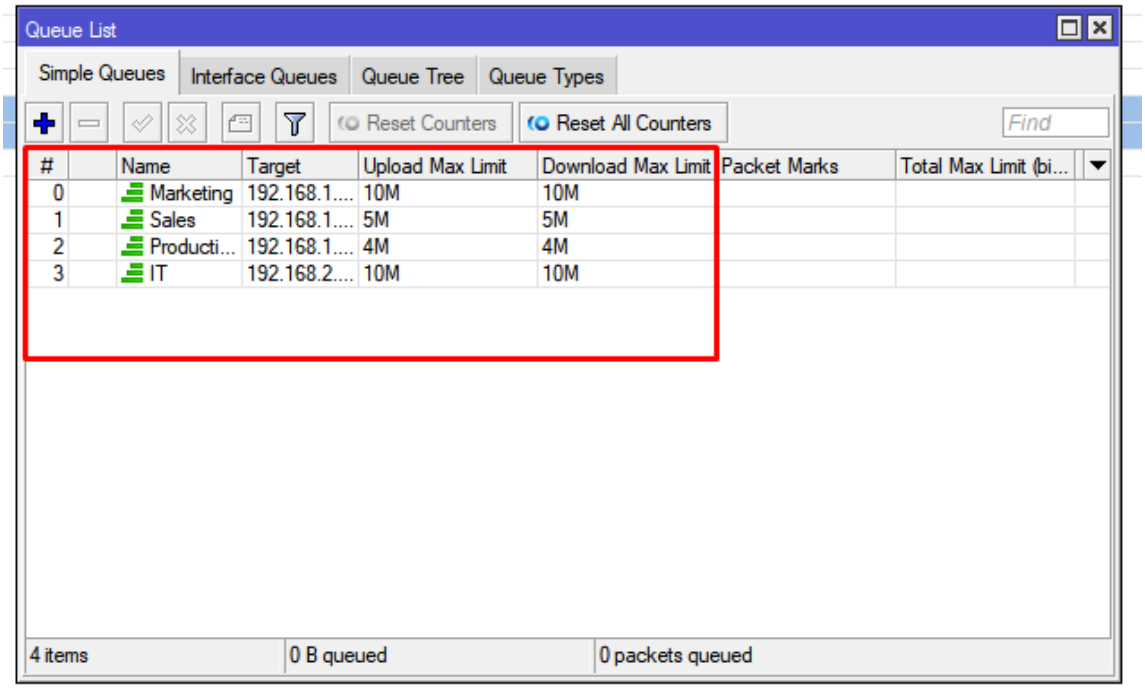




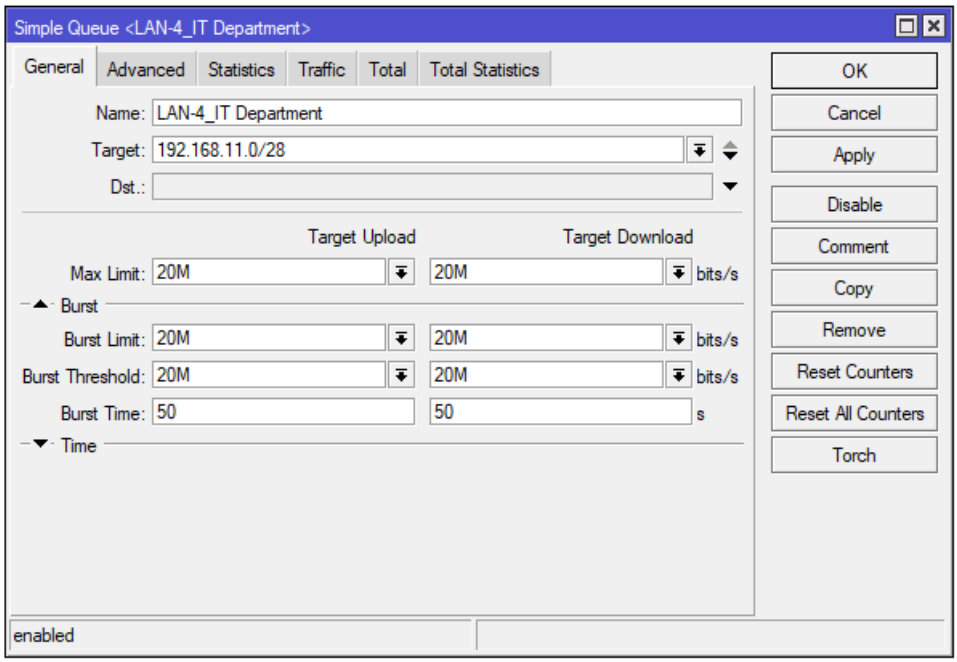
DHCP Server: DHCP LAN 1 created for Marketing department, DHCP LAN 2 created for Sales department, DHCP LAN 3 created for Production department, DHCP LAN 4 created for IT department.



Sales department get IP address 192.168.1.126 from DHCP Pool LAN 1



Distributing Bandwidth among four different LAN. 10Mbps maximum bandwidth limit for LAN 1 Marketing Department. 5Mbps maximum bandwidth limit for LAN 2 Sales Department. 4Mbps maximum bandwidth limit for LAN 3 Production Department. 10 Mbps maximum bandwidth limit for LAN 4 IT Department.



This is the Simple Queues for LAN-4 IT Department. Maximum Limits of Upload & Download Speed is 20 Mbps.